

Phase 1 – Foundations & Data Handling

1. Introduction to Python

Python is a **high-level, interpreted language** known for its readability.

- **Execution:** Code runs line-by-line via the Python Interpreter.
 - **Comments:** Use # for single-line notes to explain your logic.
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2. Variables & Constants

Variables are "containers" for storing data values.

- **Naming Rules:** Must start with a letter or underscore, cannot start with a number, and are case-sensitive (age vs Age).
- **Assignment:** Use the = operator.

```
Python
user_name = "John" # String
user_age = 25      # Integer
```

3. Data Types & Arithmetic

Common data types include:

- **int:** Whole numbers (e.g., 10, -5)
- **float:** Decimal numbers (e.g., 10.5)
- **str:** Text wrapped in quotes (e.g., "Hello")
- **bool:** Logical values (True or False)

Basic Math:

- +, -, *, / (Standard math)
 - // (Floor division: returns whole number)
 - % (Modulo: returns the remainder)
 - ** (Exponentiation: \$2**3 = 8\$)
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4. String Manipulation

Strings are sequences of characters.

- **Indexing:** Access characters by position (starts at 0).
 - name = "Python" → name[0] is 'P'.
 - **Slicing:** name[0:2] extracts 'Py'.
 - **Key Methods:**
 - .upper() / .lower(): Changes case.
 - .strip(): Removes whitespace.
 - .replace("old", "new"): Swaps text.
 - **f-Strings:** The modern way to format strings:
 - Python
 - print(f"Hello, {user_name}!")
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5. Input & Output

- **print():** Displays data to the console.
- **input():** Captures data from the user. **Note:** input() always returns a string. Use int() or float() to convert it for math.

```
Python
score = int(input("Enter your score: "))
```